



IILM University, Greater Noida

Artificial Intelligence Internship

Organization: CodeAlpha

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Enrollment No.:	2410030259
Program & Semester:	B.Tech CSE – AI/ML & V Sem
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About the Organization

CodeAlpha — AI Internship Platform

CodeAlpha is a software development and ed-tech organization offering structured, project-based virtual internship programs to students and early-career developers across India and internationally.

The AI Internship requires interns to design and build real-world AI systems, submitted as public GitHub repositories and evaluated on functionality, code quality, and documentation.

Successful completion earns an official Internship Completion Certificate and — where merited — a Letter of Recommendation.

TYPE

Virtual · Remote Internship

DOMAIN

Artificial Intelligence

MODE

Project-Based, Self-Paced

DELIVERABLE

GitHub Repos + Certificate

Internship Overview



Problem Statement	Approach	Tech Stack
Users may ask the same FAQ in different ways. A simple keyword lookup can miss paraphrased questions, so the application needs a similarity-based way to find the closest stored FAQ.	• Prepare a question-answer FAQ dataset. • Process text with NLP techniques. • Convert FAQ questions into TF-IDF vectors. • Compare questions using cosine similarity. • Return the answer linked to the best match.	• Python • NLTK • scikit-learn • TF-IDF Vectorizer • Cosine Similarity • Streamlit • Pandas / CSV

Language Translation Tool

Multilingual NLP · Google Translate API Integration

Problem Statement	Approach	Tech Stack
Language differences can make information difficult to access. The project focuses on providing a lightweight interface where users can enter text, choose a target language and receive a translation quickly.	• Accept text from the user. • Detect the input language automatically. • Let the user select the target language. • Use deep-translator for translation. • Display and copy the translated result.	• Python • Streamlit • deep-translator • Pyperclip • Multiple language support

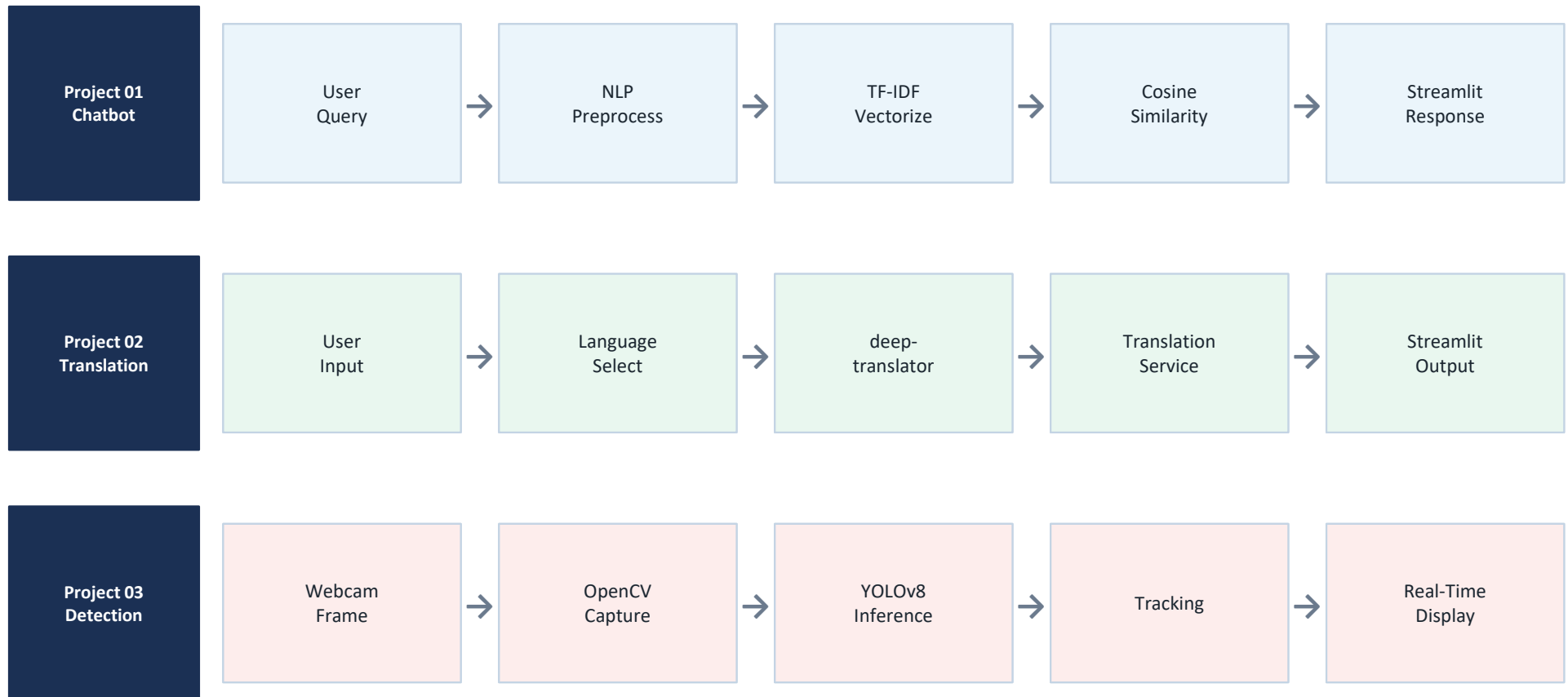
Object Detection & Tracking

Computer Vision · Real-Time YOLOv8 Inference

Problem Statement	Approach	Tech Stack
Real-time object detection requires more than identifying objects in a single image. The system must process continuous webcam frames, draw detections and maintain object identities across frames.	• Capture live frames using OpenCV. • Run YOLOv8 Nano inference. • Classify objects and draw bounding boxes. • Assign unique tracking IDs. • Display labels, confidence and counts.	• Python • OpenCV • YOLOv8n • Ultralytics • NumPy • Webcam

System Architecture

End-to-end data flow for all three projects



Technologies Used

Technology	Purpose	Project(s)
Python 3.11.09	Core programming language for all three projects	All
NLTK	Text preprocessing: tokenization, lemmatization, stopwords	Project 01
scikit-learn	TF-IDF vectorization and cosine similarity computation	Project 01
Streamlit	Interactive web UI deployment for chatbot & translation tool	Projects 01, 02
deep-translator	Python wrapper for the Google Translate API	Project 02
Ultralytics YOLOv8	Real-time object detection and multi-object tracking	Project 03
OpenCV (cv2)	Webcam frame capture, bounding box annotation and display	Project 03
Git & GitHub	Version control and project submission platform	All

Key Learnings

NLP & Machine Learning

- Built a retrieval workflow using TF-IDF and cosine similarity.
- Understood how text can be represented as numerical vectors.
- Learned the difference between retrieval-based and generative approaches.

Systems, Vision & Engineering

- Worked with live webcam frames and object detection.
- Understood detection versus tracking across consecutive frames.
- Learned how AI models are integrated into user-facing applications.
- Improved debugging, dependency and documentation skills.

Outcomes & Results

✓ All 3 Projects Accepted by CodeAlpha

All three AI projects were reviewed and accepted by CodeAlpha, meeting the functional and documentation standards of the internship program.

✓ Internship Completion Certificate

An official Completion Certificate was issued by CodeAlpha upon successful submission of all three assigned AI projects.

✓ Letter of Recommendation (LOR)

A Letter of Recommendation was awarded recognising the quality and completeness of work delivered throughout the internship.

✓ 3 Public GitHub Repositories

All projects are documented and publicly available on GitHub, forming a concrete AI/ML engineering portfolio with real deliverables.

Thank You

Akarsh Kumar

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